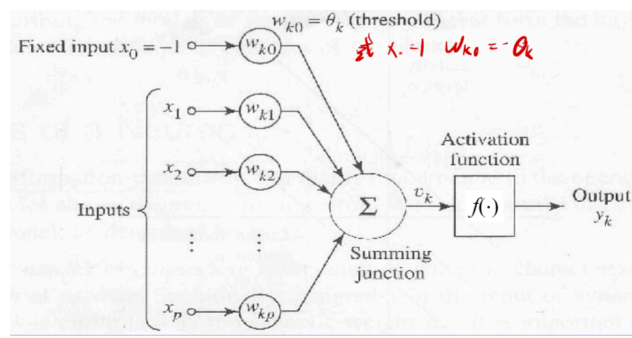
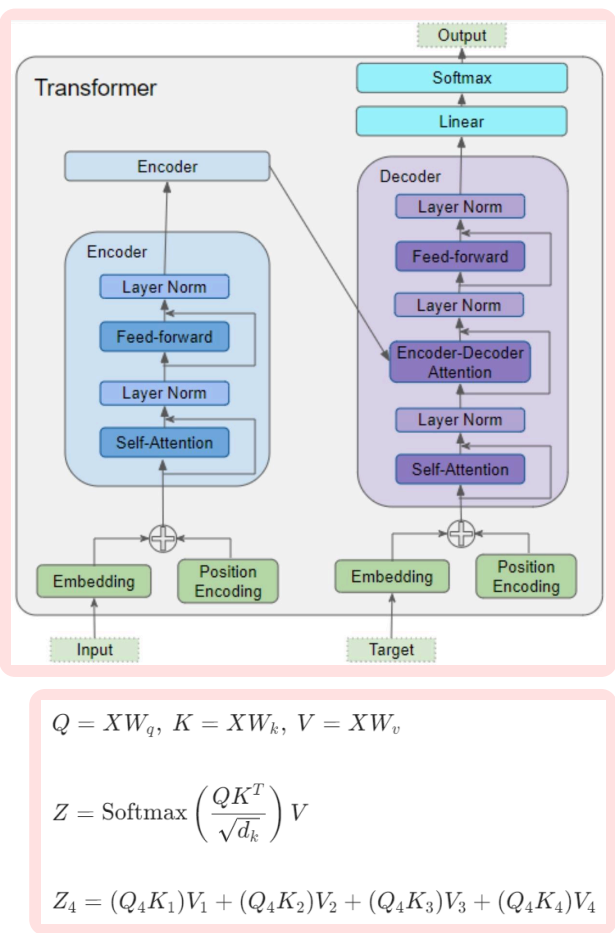


EE7403
Image Analysis and
Pattern Recognition

topic 15
Deep Learning: from CNN to Transformer



Binary Function: $f(v) = \begin{cases} 1 & v \geq 0 \\ 0 & v < 0 \end{cases}$

Sigmoid Function: $f(v) = \frac{1}{1 + \exp(-av)}$

ReLU Function: $f(v) = \max(0, v) = \begin{cases} v & \text{for } v \geq 0 \\ 0 & \text{for } v < 0 \end{cases}$

$$z_j = f\left(\sum_{i=1}^n w_{ji}x_i\right) = f(W_j^T \mathbf{x})$$

$$\mathbf{y} = f(W^T \mathbf{x}) = f(V^T f(W^T \mathbf{x}))$$

$$\mathbf{w} = (\mathbf{X}\mathbf{X}^T)^{-1}\mathbf{X}\mathbf{t}^T$$

$$v_{kj} \leftarrow v_{kj} - \eta \frac{\partial f(\mathbf{w}, \mathbf{v})}{\partial v_{kj}}$$

$$w_{ji} \leftarrow w_{ji} - \eta \frac{\partial f(\mathbf{w}, \mathbf{w})}{\partial w_{ji}}$$

$$v_{kj} \leftarrow v_{kj} + \eta(t_k - y_k)f'(t_k)z_j$$

$$w_{ji} \leftarrow w_{ji} + \eta \left[\sum_{k=1}^c v_{kj}(t_k - y_k)f'(t_k) \right] f'(t_i)x_i$$

$$y_{i,j,k} = \sum_{n=1}^c \sum_{m=1}^1 \sum_{l=1}^1 w_{l,m,n,k}^1 x_{l,j-m,n} + b_k$$

$$\mathbf{W} = (\mathbf{W}^1, \dots, \mathbf{W}^N, \dots, \mathbf{W}^P)$$
$$= \begin{pmatrix} \mathbf{R}^1 & 0 & 0 & \mathbf{R}^2 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & \mathbf{R}^1 & 0 & 0 & \mathbf{R}^2 \end{pmatrix}$$

$$S^T = \frac{1}{4} \sum_{i=1}^q (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^T = \frac{1}{q} \sum_{i=1}^q \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^T = \frac{1}{4} \mathbf{X} \mathbf{X}^T$$

$$S^T \phi = \lambda \phi$$

$$\phi^T S^T \phi = \lambda$$

$$\Sigma \Phi = \Phi \Lambda$$

$$\mathbf{y}_i = \Phi^T(\mathbf{x}_i - \mu)$$

$$\Delta = \mathbf{x} - \tilde{\mathbf{x}} = \mathbf{x} - \Phi \mathbf{y} - \mu$$

$$E[\|\Delta\|^2] = \sum_{k=q+1}^n \lambda_k$$

$$\Sigma^T = \Sigma$$

$$\phi_i \phi_j^T = \begin{bmatrix} a & b \\ b & c \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} \rightarrow \phi_i^T \phi_j = 1$$

$$d_{\mu}^2(x, \mu) = (x - \mu)^T F(x - \mu)$$

$$d_{\Sigma}^2(x, \mu) = (x - \mu)^T \Sigma^{-1}(x - \mu)$$

$$\Sigma_i = \frac{1}{q} \sum_{x_i \in \omega_i} (\mathbf{x}_i - \mu_i)(\mathbf{x}_i - \mu_i)^T$$

$$\mu_j = \frac{1}{q} \sum_{x_i \in \omega_j} \mathbf{x}_i$$

$$S^w = \sum_{j=1}^c \frac{q_j}{q} \Sigma_j$$

$$S^b = \sum_{j=1}^c \frac{q_j}{q} (\mu_j - \mu)(\mu_j - \mu)^T$$

$$\mu = \frac{1}{q} \sum_{i=1}^q \mathbf{x}_i = \sum_{j=1}^c \frac{q_j}{q} \mu_j$$

$$S^t = S^w + S^b$$

$$g_i(\mathbf{x}) = g_j(\mathbf{x})$$

$$\text{decision boundary} \quad \text{decision rule}$$

$$g_i(\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln p(\omega_i)$$

$$\text{discriminant functions}$$

$$g_i(\mathbf{x}) = \ln p(\mathbf{x}|\omega_i) + \ln p(\omega_i)$$

$$p(\mathbf{x}|\omega_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \mu_i)^T \Sigma_i^{-1}(\mathbf{x} - \mu_i)\right]$$

$$g_i(\mathbf{x}) = d_{E\omega}(\mathbf{x}, \mu_i)$$

$$\text{quadratic classifier} \quad \text{multivariate Gaussian}$$

$$g_i(\mathbf{x}) = d_{E\omega}(\mathbf{x}, \mu_i)$$

$$\text{minimum Mahalanobis distance classifier:}$$

$$\text{1. all classes own the same covariance matrix (linear classifier / hyperplane)}$$

$$\text{2. all classes own the same prior probability}$$

$$g_i(\mathbf{x}) = d_{E\omega}(\mathbf{x}, \mu_i)$$

$$\text{minimum Euclidean distance classifier:}$$

$$\text{1. all classes own the same diagonal, scalar covariance matrix (hyperplane orthogonal to the line linking the two means)}$$

$$\text{2. all classes own the same prior probability}$$

$$\text{Decide: } \omega_k = \arg \max_{\omega_i} [p(\omega_i|x)] = \arg \min_{\omega_i} [p(\omega_i|x)]$$

$$\text{maximum a posterior probability decision}$$

$$p(\omega_i|x) = \frac{p(\omega_i)p(x|\omega_i)}{p(x)}$$

$$p(x) = \sum_{i=1}^c p(x|\omega_i)p(\omega_i)$$

topic 10
Discriminant Functions and Classifiers

topic 12
Feature Extraction / Dimension
Reduction v. Machine Learning

topic 14
Neural Networks and Deep Machine
Learning: from MLP to CNN

topic 2
LSI Systems & Transforms

topic 5
Image Restoration

topic 9
Statistical Estimation & Machine Learning

topic 7
Image Analysis

topic 6
Image Analysis

topic 4
Morphological Image Processing

topic 3
Image Enhancement

topic 1
Image Fundamentals & human Perception

Image Decomposition and Linear Shift Invariant
Image Processing System

Two-dimensional Convolution and Its Properties

Two-dimensional Fourier Transform and Its Properties

Image Sampling

Introduction

Noise and Degradation Models

Inverse Filtering

Wiener Filtering

Frequency Domain Filtering

Impulse $\delta(x, y) = \begin{cases} 1, & x = y = 0 \\ 0, & \text{otherwise} \end{cases}$

$$g(x, y) = T\{f(x, y)\} = \sum_{l=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} f(l, m)h(x-l, y-m)$$

$$\Delta f(x, y) + h(x, y) = \sum_{l=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} h(l, m)f(x-l, y-m)$$

Linear and shift - invariant (LSI) T

Commutative / Associative / Distributive

Fourier transform

$$F(u, v) = \mathcal{F}\{f(x, y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \exp[-j2\pi(ux + vy)] dx dy$$

$$f(x, y) = \mathcal{F}^{-1}\{F(u, v)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) \exp[j2\pi(ux + vy)] du dv$$

$$F(u, v) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(x, y) \exp[-j2\pi(ux + vy/n)]$$

$$f(x, y) = \frac{1}{mn} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} F(u, v) \exp[j2\pi(ux + vy/n)]$$

Periodicity:

$$F(u, v) = F(u + m, v) = F(u, v + n)$$

$$f(x, y) = F(-u, -v), \quad |F(u, v)| = |F(-u, -v)|$$

Conjugate symmetry for real image $f(x, y)$

Linearity and scaling:

$$\mathcal{F}\{\alpha f_1(x, y) + \beta f_2(x, y) + \dots\} = \alpha F_1(u, v) + \beta F_2(u, v) + \dots$$

$$\mathcal{F}\{f(ax, by)\} = \frac{1}{|a|b|} F(u/a, v/b)$$

Convolution theorem:

$$f(x, y) * g(x, y) \Leftrightarrow F(u, v)G(u, v)$$

$$f(x, y)g(x, y) \Leftrightarrow F(u, v) \otimes G(u, v)$$

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